UNIVERSITY OF HERTFORDSHIRE

FINAL YEAR PROGRESS REPORT

AI-BASED INTERACTIVE SYSTEM IN HUMAN-COMPUTER INTERACTION USING NATURAL LANGUAGE PROCESSING FOR DISABLED PEOPLE

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2025/2026 ACADEMIC SESSION

1.0 Introduction

This progress report presents the development progress of the Final Year Project entitled "AIBased Interactive System in Human-Computer Interaction Using Natural Language Processing for Disabled People". The project focuses on developing an offline voice-controlled desktop automation system capable of performing various computer operations through speech commands. The system aims to improve accessibility and provide hands-free interaction for users, particularly individuals with physical disabilities and mobility limitations.

2.0 Progress Achieved

Several major project activities have been successfully completed throughout the project duration.

2.1 Literature Review

A comprehensive literature review was conducted to investigate Human-Computer Interaction (HCI), Artificial Intelligence (AI), Natural Language Processing (NLP), Automatic Speech Recognition (ASR), and desktop automation technologies. The review included the analysis of VOSK, PocketSphinx, and CMUSphinx speech-recognition frameworks.

2.2 System Design and Methodology

The project methodology, system architecture, workflow design, use case analysis, and testing procedures were successfully designed and documented. A prototype-based development methodology was adopted to guide the implementation process.

2.3 System Development

The offline voice-controlled desktop automation system was successfully implemented using Python. The system integrates speech recognition with desktop automation technologies to perform tasks such as:

- Mouse movement control
- Scrolling operations
- Browser automation
- Calculator control
- Camera access
- Volume management
- Application launching

2.4 System Testing and Evaluation

Comprehensive testing was conducted to evaluate system performance. Functional testing, command execution validation, Word Error Rate (WER) analysis, runtime monitoring, environmental noise testing, and user evaluation involving 17 participants were successfully completed.

2.5 Documentation and Deliverables
The final report, poster, presentation slides, GitHub repository, project files, and video demonstration have been successfully completed and submitted.

3.0 Current Project Status
The project has been completed successfully. All project objectives have been achieved and all required deliverables have been prepared for final assessment and submission.

Project Completion Status: 100%

4.0 Challenges Encountered
Several challenges were encountered during the development process, including speech recognition inaccuracies under noisy environments, command interpretation issues, and integration challenges between speech-recognition frameworks and desktop automation modules. These issues were addressed through testing, command validation mechanisms, and optimisation of speech-recognition settings.

5.0 Future Improvements
Future enhancements may include:

- Multilingual speech recognition support.
- Conversational AI capabilities.
- Speaker authentication features.
- Cross-platform compatibility.
- Deep learning-based speech recognition models.
- Smart home and IoT integration.

6.0 Conclusion
Overall, the project has achieved its objectives by successfully developing an offline voice-controlled desktop automation system. The system demonstrates that speech recognition technologies can effectively support hands-free Human-Computer Interaction while maintaining privacy, accessibility, and operational reliability.

RESEARCH STUDENT PROGRESS LOGBOOK

Project Title: AI-Based Interactive System in Human-Computer Interaction Using Natural Language Processing for Disabled People

Student Name: Mohammad

Student ID: SCSJ2000064

Supervisor: Ms. Diana

Date & Day	Time & Location	Duration	Discussion with Supervisor	Supervisor Comments	Planned Action	Supervisor Signature
03/10/2025 Friday	10:00 AM – 10:30 AM, Level 11 Lecturer Room	30 Mins	Discussed FYP topic selection. Proposed AI-based voice-controlled system for disabled users. Discussed project scope and expected outcomes.	Refine project objectives. Focus on offline speech recognition.	Prepare proposal draft. Conduct literature review.	
31/10/2025 Friday	10:00 AM – 10:45 AM, Level 11 Lecturer Room	45 Mins	Presented proposal. Discussed problem statement, research questions, objectives and project timeline.	Strengthen literature review. Add more recent references.	Revise proposal. Continue literature review.	
21/11/2025 Friday	10:00 AM – 10:45 AM, Student Discussion Area, Level 11	45 Mins	Presented literature review findings. Discussed AI, HCI and speech recognition technologies. Compared VOSK and PocketSphinx.	Improve framework comparison. Expand discussion of related works.	Complete Chapter 2. Begin methodology development.	
05/12/2025 Friday	11:00 AM – 11:45 AM, Waiting Area,	45 Mins	Presented methodology and system design. Discussed workflow, architecture and testing methods. Reviewed progress before semester	Improve system diagrams. Clarify evaluation process.	Finalise methodology chapter. Prepare	

	Level 11		break.		implementation plan.	
Semester break	-	-	December 2025 – Mid March 2026	-	-	-
16/03/2026 Monday	2:00 PM – 2:45 PM, Level 11 Lecturer Room	45 Mins	Resumed project after semester break. Reviewed implementation plan and milestones.	Begin prototype implementation. Focus on command recognition reliability.	Develop first working prototype.	
27/03/2026 Friday	11:00 AM – 11:45 AM, Student Discussion Area, Level 11	45 Mins	Demonstrated first prototype using VOSK speech recognition.	Improve recognition accuracy. Add desktop automation functions.	Continue implementation. Integrate more commands.	
17/04/2026 Friday	3:00 PM – 3:45 PM, Waiting Area, Level 11	45 Mins	Presented functional testing results and framework comparison.	Include WER analysis and environmental noise testing.	Continue performance evaluation and collect results.	
08/05/2026 Friday	10:00 AM – 10:45 AM, Student Discussion Area, Level 11	45 Mins	Presented VOSK, PocketSphinx and CMUSphinx evaluation results and user testing plan.	Increase participant diversity. Strengthen discussion of findings.	Conduct user testing and analyse data.	
21/05/2026 Thursday	2:00 PM – 3:00 PM, Level 11 Lecturer Room	60 Minutes	Presented completed system, user evaluation involving 17 participants and draft report.	Improve report formatting and conclusion section.	Complete corrections and prepare poster and	

					slides.	
08/06/2026 Monday	10:00 AM – 10:45 AM, Level 11 Lecturer Room	45 Minutes	Submitted final report, poster, presentation slides, project files, GitHub repository and video demonstration.	Ready for final submission.	Submit all deliverables. Project completed.	

Student Signature:

Supervisor Signature: _______________